

VPM Decompression Knowledge Base

For LSP D-Planner — Algorithm Reference & Implementation Guide

Compiled from primary sources: HHS Software MultiDeco docs, Eric Maiken's VPMechanics series, Erik C. Baker's FORTRAN reference, scubatechphilippines.com articles, and thetheoreticaldiver.org.

1. Historical Development

1974–1984 · University of Hawaii (Yount et al.)

- David Yount, D.C. Hoffman, T.D. Kunkle and others published 15+ research papers on bubble formation, cavitation nuclei, and gas stabilization in biological fluids.
- Key 1979 paper: "Skins of Varying Permeability" — the foundation of the model name. Bubble nuclei are stabilized by surfactant skins whose permeability to gas varies with pressure.
- 1984 paper: "Decompression theory: A dynamic critical-volume hypothesis" — first formal statement of the CVA concept.
- 1986 paper (Yount & Hoffman): "On the use of a bubble formation model to calculate diving tables" — first published VPM algorithm, the reference baseline for all subsequent implementations.

1992–1995 · Eric Maiken

- Began implementing the Yount/Hoffman algorithm in BASIC, then Mathematica.
- Extended to multi-level dives, nitrox mixes, and later trimix.
- Distributed code freely to researchers; became the active development centre of VPM.
- Presented "Bubble Decompression Strategies" at tek95 conference.

1998–2000 · DecoList / VPM List

- Rob Murray started the VPM List (later DecoList), an open mailing list for VPM development.
- Erik Baker (creator of Gradient Factors) joined in 1999.
- David Yount joined discussions — pleased to see his research in use.
- Over 200 members including researchers, tech divers, and industry leaders.
- Repetitive diving algorithm, multi-gas, altitude, and reverse-profile handling all developed here.

2000 · Algorithm Finalisation

- Core VPM algorithm finalised June 2000 when Yount presented his multi-dive algorithm at UHMS Stockholm.
- Key advancements over 1986 paper: multiple inert gases, dynamic critical radius (ascent/descent influence), multi-dive tracking.
- David Yount died April 27, 2000, age 64.

2001 · VPM (original)

- Erik Baker released final FORTRAN source code (public domain: "Distribute freely — credit the authors").
- Ross Hemingway adapted it to Windows as **V-Planner** (freeware, July 2002) — first public GUI VPM program.

2002–2003 · VPM-B

- Diver feedback showed the original VPM was "range bound" on deeper dives.
- Baker developed the **Boyle's Law Adjustment** (VPM-B): expands bubble radius at each stop according to Boyle's law during ascent, adding systematic conservatism for deeper dives.
- VPM-B released at NAUI Deep Stop conference, February 2003.

2005 · VPM-B/E

- VPM-B/E ("Extra") introduced for extreme/long dives (>100 min deco).
- Profile diverges from VPM-B and approaches an overlap of VPM-B + Bühlmann, for maximum safety margin.
- Dave Shaw used VPM-B for his 270m world record CCR dive at Boesmansgat.

2011 · VPM-B/FBO

- "Fast Bail Out" variant for CCR divers in OC bailout situations.
- Reduces deep stop time, expands shallow stops — 30–50% less diluent required vs standard VPM-B.
- Still more conservative than raw Bühlmann for the same profile.

2018

- MultiDeco-TDC added to TDC-3 dive computer.
- V-Planner/MultiDeco database exceeds 140,000 logged dives.

2. Physical Foundation — The Yount Bubble

2.1 Bubble Nucleus Model

VPM enhances Bühlmann's 16-compartment dissolved gas model by adding **one bubble nucleus** to each tissue compartment (32 total for trimix: separate N₂ and He bubbles per compartment).

The bubble has these properties:

- **Permeable skin:** gas from surrounding tissue can cross into the bubble when tissue tension > bubble pressure.
- **Elasticity and memory:** a compressed bubble will rebound toward its original size over a time scale of weeks (governed by `Regeneration_Time_Constant`). Negligible for normal dives; relevant for saturation.
- **Variable permeability:** below the "Gradient Onset of Impermeability" threshold, the bubble becomes impermeable — further crushing is harder.

2.2 Bubble Pressure Equation

$$P_b = P_a + S / r$$

- P_b = gas pressure inside bubble (atm)
- P_a = ambient pressure (atm)
- S = bubble skin surface tension constant (derived from `Surface_Tension_Gamma` and `Skin_Compression_GammaC`)
- r = bubble radius (metres, microns in settings)

Implication: bubbles tend to collapse because internal pressure > ambient, driving gas outward. But if tissue tension (PTC) > P_b , dissolved gas flows *into* the bubble, inflating it. This linkage between Bühlmann's dissolved gas model and Yount's bubble model is the core mechanism.

2.3 Critical Radii Parameters (Baker FORTRAN defaults)

- `Critical_Radius_N2_Microns`: valid range 0.2–1.35 μm . Default ~0.55 μm .
- `Critical_Radius_He_Microns`: valid range 0.2–1.35 μm . Default ~0.45 μm .
- Larger radius → more conservative (lower allowed supersaturation).
- MultiDeco conservatism slider increases critical radii: +1=5%, +2=12%, +3=22%, +4=35%, +5=50%.

3. The No-Bubble-Growth Criterion

Condition to prevent bubble growth:

$$PTC < P_b = P_a + S/r$$

Rearranged as supersaturation form:

$$PTC - P_a < S/r$$

The left side is tissue supersaturation. The right side is the VPM-derived maximum allowed supersaturation — a function of bubble radius only. This replaces Bühlmann's empirical M-values with physics-based limits.

Key insight: unlike Bühlmann M-values (empirically derived from DCS incidence data), VPM limits are derived from bubble mechanics. They are dive-profile-specific — the allowed supersaturation after a given dive depends on the actual crushing history of the bubbles during descent.

4. Bubble Crushing During Descent

4.1 Crushing Mechanism

Crushing is driven by $P_a - PTC$ (ambient pressure minus tissue tension), **not** simply by depth.

- Fast compartments equilibrate with ambient quickly $\rightarrow$ small $P_a - PTC \rightarrow$ less crushing.
- Slow compartments lag behind $\rightarrow$ large $P_a - PTC \rightarrow$ more crushing $\rightarrow$ more conservative allowed tension on ascent.

4.2 Descent Rate Effects

Faster descent = more conservative ascent schedule: faster descent means fast compartments have less time to equilibrate, maximising crushing pressure differential. More crushing = smaller bubble radius = higher allowed supersaturation.

Slower descent = less crushing in fast compartments, resulting in more deco required. This is the opposite of Bühlmann behaviour.

4.3 Reverse Profile Implications

A dive to 150 ft then 300 ft produces less bubble crushing in fast compartments than an immediate 300 ft descent. Therefore reverse profiles genuinely require more conservatism under VPM — this is a physics-based prediction, not an empirical rule.

4.4 Impermeability Threshold

Once $P_a - PTC$ exceeds `Gradient_Onset_of_Imperm_Atm`, the bubble becomes impermeable. Further depth increases do not proportionally reduce radius — the benefit of deeper descent is diminished beyond this point.

5. The Critical Volume Algorithm (CVA)

Using no-bubble-growth alone produces overly conservative profiles near the NDL. The CVA relaxes this by permitting controlled bubble growth up to a critical volume threshold.

5.1 How CVA Works

1. Run VPM profile using strict no-bubble-growth → produces initial conservative schedule.
2. Relax the tissue tension limit slightly → compute a new schedule.
3. Estimate excess bubble volume = $\int(\text{excess supersaturation} \times \text{time})$ across all compartments.
4. If volume < Critical_Volume_Lambda, tighten the limits and recompute.
5. Iterate until the schedule converges (phase volume time change < 1 min between iterations — typically 4 iterations).

5.2 CVA Parameter

- Crit_Volume_Parameter_Lambda: default 7500 fsw·min (Bruce Wienke). Adjustable 6500–8300.
- Lower lambda → more conservative (less bubble volume allowed).
- The CVA is always on by default. Can be disabled in program settings.

5.3 Why CVA Matters for Implementation

The iterative loop is computationally heavier than Bühlmann. Each pass recomputes the full ascent. In LSP D-Planner's JS engine (window.VPMEngine), this loop is why VPM-B takes longer to compute than ZHL-GF.

6. VPM-B: Boyle's Law Adjustment

6.1 Mechanism

After the initial deco ceiling is established:

1. At the **first deco stop**, note bubble radius in each compartment.
2. At each **subsequent stop**, increase radius by Boyle's law: $r_{\text{new}} = r_{\text{old}} \times (P_{\text{old}} / P_{\text{new}})^{(1/3)}$ (volume expansion → radius increase).
3. Larger radius → lower S/r → lower allowed supersaturation → diver stays longer.

6.2 Effect on Profiles

- Minimal impact near NDL (short, shallow dives).
- Major impact on deeper/longer dives — can add 30–80+ minutes to very deep trimix profiles.
- Produces characteristic VPM-B shape: moderate deep stops + somewhat reduced shallow time vs GF 30/85, but with physically-motivated stop depths.

6.3 GF Equivalence (from thetheoreticaldiver.org analysis)

For a 120m TMX dive (20 min bottom), VPM-B+1 yields effective gradient factors of:

- ~40–60% GF equivalent at deep stops
- ~124% GF equivalent at last stop (3m) for the leading tissue
- ~180% for slow non-leading tissues (these never control the ascent)

Approximate GF equivalent: GF 20/125 produces a profile very similar to VPM-B+1 for deep trimix. This is why VPM-B allows higher supersaturation near the surface — slow tissue saturation there is low.

For LSP D-Planner VPM-B/GF hybrid (VPMB_GFS): this data supports using a high GF-hi value (85–100) as the Bühlmann surfacing constraint layered over VPM-B ceilings. VPM-B naturally surfaces "hot" for slow tissues; GFS caps this.

7. Compartment Half-Times (Baker FORTRAN canonical values)

N₂ Half-Times (ZHL-16, 16 compartments with 1b variant)

```
5.0, 8.0, 12.5, 18.5, 27.0, 38.3, 54.3, 77.0,  
109.0, 146.0, 187.0, 239.0, 305.0, 390.0, 498.0, 635.0 [minutes]
```

He Half-Times

```
1.88, 3.02, 4.72, 6.99, 10.21, 14.48, 20.53, 29.11,  
41.20, 55.19, 70.69, 90.34, 115.29, 147.42, 188.24, 240.03 [minutes]
```

Note: Maiken's VPMechanics notebooks used a slightly different symmetric ZHL-16 set:

{4, 8, 12.5, 18.5, 27, 38.3, 64.3, 77, 109, 146, 187, 239, 305, 390, 498, 635} — compartment 7 differs (64.3 vs 54.3). Baker's is the standard for VPM-B.

8. M-Values vs VPM Limits

8.1 Bühlmann ZHL-16C M-Values

Each compartment has a linear M-value equation:

$$M(P_{\text{amb}}) = a + P_{\text{amb}} / b$$

Where a and b are per-compartment, per-gas coefficients. For mixed gas:

$$\begin{aligned} a_{\text{mix}} &= (a_{\text{He}} \times P_{\text{He}} + a_{\text{N}_2} \times P_{\text{N}_2}) / (P_{\text{He}} + P_{\text{N}_2}) \\ b_{\text{mix}} &= (b_{\text{He}} \times P_{\text{He}} + b_{\text{N}_2} \times P_{\text{N}_2}) / (P_{\text{He}} + P_{\text{N}_2}) \end{aligned}$$

ZHL-16B is used for table calculations; ZHL-16C (more conservative at middle compartments) is for real-time computers. LSP D-Planner uses ZHL-16C (correct for a live calculator).

8.2 Workman vs Bühlmann Format

- **Workman** (US Navy): M_0 (surfacing value) + ΔM (slope) based on depth pressure (gauge).
- **Bühlmann**: coefficients a (intercept at zero absolute pressure) and b (reciprocal slope), based on absolute pressure — designed for altitude diving.

Conversion: $M_{0_workman} = a + 1/b$ (at 1 atm absolute). Bühlmann's formulation is correct for altitude; LSP D-Planner's altitude pressure correction to Bühlmann a/b coefficients is therefore the right approach.

8.3 VPM "vpM-Values"

VPM produces dynamic, profile-dependent limits that replace Bühlmann M-values. Key differences:

- VPM limits are **tighter at depth** (more deep stops) and **looser shallow** (less shallow time).
- From Maiken VPMechanics5: VPM M-values cross below Bühlmann at depth (producing deep stops), then exceed Bühlmann for slow compartments near the surface (producing less shallow time for a given dive profile).
- Both models ultimately stage divers by limiting dissolved gas tension — same paradigm, different limit derivation.

9. MultiDeco Implementation Notes

9.1 Gas Switching Logic

MultiDeco controls deco gas switches by ppO_2 limits set in configuration. A zero-time leg (time = -) triggers a gas switch:

```
25, -, 32      (switch to 32% EAN at 25m)
65, 30, 12/60  (30 min at 65m on 12/60 TMX)
```

Deco gases are not used until all bottom-plan legs are processed.

Implication for LSP D-Planner: our gas switching by MOD threshold is consistent with this approach. The travel gas zero-time leg maps to our `#travelGasSwitch` depth concept.

9.2 Conservatism in MultiDeco

MultiDeco conservatism levels increase critical radii:

Level	Critical Radius Increase
Nominal	baseline

Level	Critical Radius Increase
+1	5%
+2	12%
+3	22%
+4	35%
+5	50%

Warning from MultiDeco docs: Do NOT fake conservatism by inflating depth/time in VPM. This reduces the repetitive dive penalty — subsequent dives will have insufficient deco time.

9.3 Run Time vs Stop Time

MultiDeco displays **run times**, not stop times. Stop times include transit time between stops. **Follow run times, not stop times.**

9.4 Deep Stop Warnings

From MultiDeco FAQ:

- Do not add extra stops in deep sections — significantly increases total deco obligation.
- Extending shallow stops (6m, 3m) has minimal side effects.
- Ascent rate deviations matter most in the **deep portion** of ascent.
- If forced to use new gas (lost deco mix): wait the time difference between the two plans at current depth, then follow new plan timings.

9.5 Deco Gas Acceleration

VPM calculates in an iterative process. Having deco mixes available can shorten mid-level stops **even before the switch point** because the algorithm accounts for future off-gassing capability when generating the initial ceiling schedule.

10. Altitude Diving (VPM Context)

From Baker's FORTRAN:

```
IF (Altitude_Dive_Algorithm_Off) THEN
  Altitude_of_Dive = 0.0
  CALL CALC_BAROMETRIC_PRESSURE (Altitude_of_Dive)
  ! Initial N2 pressure = (Baro - WaterVapor) × 0.79
ELSE
  CALL VPM_ALTITUDE_DIVE_ALGORITHM
END IF
```

The altitude algorithm adjusts:

1. **Barometric pressure** (affects ambient pressure throughout)
2. **Initial critical radii** (bubbles are larger at altitude due to lower ambient pressure — more conservative ascent required)
3. **Initial tissue N₂ loading** (already partially off-gassed if diver ascended to altitude before dive)

LSP D-Planner altitude implementation uses `altSurfaceP` (bar) for all pressure calculations. The critical radii adjustment for altitude is not currently implemented — this is a known conservative simplification (our VPM will be slightly under-conservative at altitude since we don't increase critical radii for the lower initial ambient pressure).

11. Repetitive Dives

From Baker FORTRAN: the repetitive dive loop saves tissue loadings and bubble state at surface, accounts for surface interval off-gassing, then carries forward adjusted critical radii for the next dive.

Key rule: **critical radii are NOT reset between dives**. The bubble crushing from dive 1 modifies the effective critical radius that constrains dive 2's ascent.

From VPMechanics10:

- Inter-dive surface interval allows tissue off-gassing per normal Bühlmann kinetics.
- Bubble regeneration over the surface interval (weeks time scale) is negligible for sport/tech diving.
- Residual bubble state from dive 1 makes dive 2 constraints more conservative than an identical single dive.

LSP D-Planner multi-dive implementation (`switchTab('multi')`) carries tissue pressure state between dives but should also carry VPM bubble state. Currently the engine resets critical radii between repetitive dive segments — noted for future improvement.

12. Water Vapour & Other Gas Pressures

Baker FORTRAN (Schreiner value, respiratory quotient 0.8):

- FSW: `Water_Vapor_Pressure = 1.607 fsw → 0.0627 bar`
- MSW: `Water_Vapor_Pressure = 0.493 msw → 0.0627 bar`

Alveolar N₂ partial pressure = $(\text{Barometric_P} - \text{Water_Vapor_P}) \times 0.79$

LSP D-Planner uses `WATER_VAPOUR = 0.0627 bar` — consistent with Baker's canonical value.

`Constant_Pressure_Other_Gases` (CO₂ + O₂ treated as constant background):

- Default: `Pressure_Other_Gases_mmHg / 760 × Units_Factor`
- Typically ~0.02 bar

13. VPM-B/GF Hybrid (VPMB_GFS in LSP D-Planner)

From MultiDeco docs: "VPM-B + GFS takes a regular VPM-B profile and adds a second calculation from ZHL-16 GF profile with the GF-Hi value specified. From these two, the most restrictive ceiling/time is used."

Implementation logic:

```
ceiling = max(VPM-B ceiling, Bühlmann GF-Hi ceiling)
stop_time = max required to satisfy both models
```

GFS "normally only takes effect on larger dives" and "adds a few minutes to the last few stops."

Practical note: for a 120m trimix dive, VPM-B at the 3m stop has an effective GF equivalent of ~124% (leading tissue). With GFS GF-Hi = 85%, Bühlmann ceiling at 3m would be much lower — GFS adds significant time at the last 1–2 stops of very deep dives. For shallower dives (<50m), GFS has negligible effect.

14. Key VPM Parameters Quick Reference

Parameter	Typical Value	Effect
Critical_Radius_N2	0.55 μm	Larger = more conservative
Critical_Radius_He	0.45 μm	Larger = more conservative
Surface_Tension_Gamma	17.9 dyn/cm	Skin surface tension
Skin_Compression_GammaC	257 dyn/cm	Skin compression constant
Crit_Volume_Lambda	7500 fsw·min	CVA threshold (lower = more conservative)
Gradient_Onset_Imperm	8.2 atm	Depth at which bubble becomes impermeable
Regeneration_Time_Constant	20160 min (~2 weeks)	Bubble size recovery rate
Water_Vapor_Pressure	0.493 msw / 1.607 fsw	Schreiner value, RQ=0.8
Pressure_Other_Gases	~5.32 mmHg	Background CO ₂ /O ₂

15. Open Source Implementations

Source	Language	Notes
Erik Baker FORTRAN	FORTRAN 77	Canonical reference, public domain
bwaite/vpmb	Python	Clean port of Baker's FORTRAN
Subsurface	C++	Full dive log + VPM-B planning
LSP D-Planner VpmbEngine.java	Java	Our server-side reference
LSP D-Planner window.VPMEngine	JavaScript	In-browser implementation

The Baker FORTRAN is the ground truth. Any discrepancy between LSP D-Planner output and Baker's FORTRAN output is a bug in our implementation.

16. Profile Comparison: VPM-B vs GF 30/85 (120m TMX dive)

From thetheoreticaldiver.org (120m, 20 min BT, TMX18/50, EAN50, EAN100):

Model	First Stop	Surface Runtime
Bühlmann GF 30/80	51m	188 min
VPM-B+1	63m	141 min
Bühlmann GF 20/125	54m	135 min

VPM-B produces:

- Deeper first stop (63m vs 51m)
- Less total time (141 vs 188 min)
- Shorter shallow stops (22 min at 3m vs 40 min for GF 30/80)
- More time distributed across deep stops

GF 20/125 approximates VPM-B+1 closely for this profile type.

17. Practical Diver Notes (MultiDeco FAQ)

1. Follow run times. Stop times include transit between stops.
2. No extra deep stops. Adding deep stops significantly increases total obligation.
3. Extending shallow stops (6m, 3m) is safe.
4. Ascent rate deviations are most critical in the deep deco zone.
5. Lost deco gas: wait the time difference at current depth, then follow new plan.
6. Conservatism: use +2 or +3 for most divers. +3 or +4 for cold, strenuous, multi-day, or prior DCS history.
7. Do not fake VPM conservatism by inflating depth/time — this breaks the repetitive dive calculations.
8. High O₂ deco mixes after trimix generally do not accelerate deco (helium deco mixes close to the switch point do accelerate it).
9. First deco gas switch works best when at or just above the first deco stop depth.

Sources

- [MultiDeco FAQ](#) — HHS Software
- [The VPM Model \(History & Variants\)](#) — HHS Software
- [VPM References](#) — HHS Software
- [VPM Algorithm Details](#) — Eric Maiken
- [VPMechanics1](#) — Eric Maiken (100fsw air dive, iterative convergence)
- [VPMechanics5](#) — Eric Maiken (VPM M-values vs Bühlmann, hypobaric/hyperbaric)
- [VPM for Dummies](#) — Kevin Watts / ScubaTech Philippines
- [Understanding M-Values](#) — Erik C. Baker
- [VPM-B Gradients as Gradient Factors](#) — The Theoretical Diver
- [Baker FORTRAN VPM-B Source](#) — Erik C. Baker / Eskill Gist